

Parameter Estimation of a Rotated Exponential Sinusoidal Curve

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1 Introduction

The objective of this assignment is to estimate the unknown parameters of a parametric curve using only the observed (x, y) coordinates. The given curve is represented by

$$x = t \cos \theta - e^{Mt} \sin(0.3t) \sin \theta + X \quad (1)$$

$$y = 42 + t \sin \theta + e^{Mt} \sin(0.3t) \cos \theta \quad (2)$$

where

$$6 < t < 60.$$

The unknown parameters are

$$\theta, X, M.$$

Instead of directly applying nonlinear optimization, the geometric structure of the equations was analyzed first. This made it possible to recover most of the unknown parameters analytically before performing any numerical estimation.

2 Initial Observation

Ignoring the oscillatory component,

$$e^{Mt} \sin(0.3t),$$

the equations reduce to

$$x = t \cos \theta + X$$

and

$$y = 42 + t \sin \theta.$$

These equations represent a straight line rotated by an angle θ . The exponential sinusoidal term introduces oscillations around this line, while the parameter X translates the entire curve horizontally.

Based on this observation, the estimation process was divided into three stages.

1. Estimation of the rotation angle.
2. Recovery of the translation.
3. Estimation of the exponential growth parameter.

3 Estimating the Rotation Angle

Since the dominant structure of the point cloud is approximately linear, Principal Component Analysis (PCA) was used to estimate its orientation [1].

Let the first principal component be

$$\mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}.$$

The corresponding rotation angle is

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right)$$

The first principal component represents the direction of maximum variance and therefore provides a reliable approximation of the rotation angle.

4 Coordinate Transformation

Although an estimate of θ is now available, the remaining unknown parameters are still coupled together in the original equations.

To simplify the model, every point was projected onto the principal direction and its perpendicular using an orthogonal coordinate transformation [2].

The transformed coordinates are

$$u = x \cos \theta + y \sin \theta$$

and

$$v = -x \sin \theta + y \cos \theta.$$

Substituting the original equations into u gives

$$u = (t \cos \theta - e^{Mt} \sin(0.3t) \sin \theta + X) \cos \theta \\ + (42 + t \sin \theta + e^{Mt} \sin(0.3t) \cos \theta) \sin \theta.$$

Expanding,

$$u = t(\cos^2 \theta + \sin^2 \theta) \\ - e^{Mt} \sin(0.3t) \sin \theta \cos \theta \\ + e^{Mt} \sin(0.3t) \cos \theta \sin \theta \\ + X \cos \theta + 42 \sin \theta.$$

Using the trigonometric identity

$$\cos^2 \theta + \sin^2 \theta = 1,$$

the oscillatory terms cancel exactly, resulting in

$$u = t + X \cos \theta + 42 \sin \theta.$$

Let

$$C = X \cos \theta + 42 \sin \theta.$$

The transformed equation becomes

$$\boxed{u = t + C.}$$

This transformation completely removes the oscillatory component from the projected coordinate.

Applying the same transformation to the perpendicular direction gives

$$v = 42 \cos \theta - X \sin \theta + e^{Mt} \sin(0.3t).$$

Defining

$$D = 42 \cos \theta - X \sin \theta,$$

the equation simplifies to

$$\boxed{v = D + e^{Mt} \sin(0.3t).}$$

The original coupled equations are therefore separated into a linear component and an oscillatory component, considerably simplifying the parameter estimation process.

5 Recovering the Unknown Parameters

Since

$$u = t + C,$$

and the parameter is bounded by

$$6 < t < 60,$$

the constant C is estimated using the end points of the projected coordinate,

$$C = \frac{(u_{\min} - 6) + (u_{\max} - 60)}{2}.$$

Once C is known, the hidden parameter is recovered as

$$t = u - C.$$

The recovered values satisfy the expected interval and therefore provide a good approximation of the original parameterization.

The perpendicular projection contains only a constant offset and the oscillatory component,

$$v = D + e^{Mt} \sin(0.3t),$$

where

$$D = 42 \cos \theta - X \sin \theta.$$

The constant offset is estimated as

$$D = \frac{1}{N} \sum_{i=1}^N v_i.$$

The translation parameter is then obtained analytically as

$$X = \frac{42 \cos \theta - D}{\sin \theta}.$$

After estimating θ , t , and X , only the exponential growth parameter M remains unknown.

Instead of performing a three-dimensional optimization, a one-dimensional search was carried out over the interval

$$-0.05 \leq M \leq 0.05.$$

For every candidate value of M , the oscillatory component

$$e^{Mt} \sin(0.3t)$$

was reconstructed and compared with the transformed observations. The value of M corresponding to the minimum reconstruction error was selected. Since only one parameter

remained unknown, this search was computationally inexpensive while producing an accurate estimate.

6 Results

The proposed methodology produced the following parameter estimates:

$$\theta = 0.497125 \text{ rad},$$

$$X = 53.739229,$$

$$M = 0.0268749.$$

The recovered parameter interval was

$$t_{\min} = 5.961, \quad t_{\max} = 60.039,$$

which closely matches the expected interval

$$6 < t < 60.$$

The reconstructed curve achieved a mean L1 distance of

$$25.230127,$$

while the total L1 distance between the reconstructed and expected curves was

$$37845.1907.$$

These results demonstrate that the proposed analytical approach provides accurate estimates of the unknown parameters while requiring only a simple one-dimensional numerical search.

7 Conclusion

The inverse parameter estimation problem was simplified by exploiting the geometric structure of the given equations. Principal Component Analysis was used to estimate the rotation angle, after which an orthogonal coordinate transformation separated the linear and oscillatory components of the curve. The transformed equations enabled analytical estimation of the hidden parameter t and the translation parameter X , leaving only the exponential growth parameter M to be determined numerically.

Compared with solving the complete nonlinear optimization problem directly, this approach significantly reduced the complexity of the estimation process while maintaining good reconstruction accuracy.

References

- [1] I. T. Jolliffe, *Principal Component Analysis*, 2nd Edition, Springer, 2002.
- [2] Gilbert Strang, *Introduction to Linear Algebra*, 5th Edition, Wellesley-Cambridge Press, 2016.
- [3] Jorge Nocedal and Stephen J. Wright, *Numerical Optimization*, 2nd Edition, Springer, 2006.